

1. 判断下列级数是否收敛? 条件收敛还是绝对收敛?

(1) $\frac{1}{2^2} - \frac{1}{4^2} + \frac{1}{6^2} + \dots + (-1)^{n-1} \frac{1}{(2n)^2} + \dots;$

(1) $u_n = \frac{1}{(2n)^2}, \lim_{n \rightarrow \infty} u_n = 0$, 单减, 由莱布尼茨判别法, 级数收敛.

Σ 绝对收敛

(2) $1 - \frac{1}{3^p} + \frac{1}{5^p} + \dots + (-1)^{n+1} \frac{1}{(2n-1)^p} + \dots;$

(3) $\frac{1}{2\ln 2} - \frac{1}{3\ln 3} + \frac{1}{4\ln 4} + \dots + (-1)^{n-1} \frac{1}{(n+1)\ln(n+1)} + \dots;$

(4) $\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}-1}{n};$

(5) $\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \dots + (-1)^{n-1} \frac{2n+1}{n(n+1)} + \dots;$

(6) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n!}{3^{n^2}};$

(7) $\frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{3} \sin \frac{\pi}{3} + \frac{1}{4} \sin \frac{\pi}{4} - \dots;$

(8) $\sum_{n=1}^{\infty} (-1)^{n+1} \tan \frac{\varphi}{n} \quad \left(-\frac{\pi}{2} < \varphi < \frac{\pi}{2}\right);$

(3) $\lim_{n \rightarrow \infty} u_n = 0$, $f(n) = (n+1)\ln(n+1)$, $f'(n) = \ln(n+1) + 1 > 0$, 故 $\{u_n\}$ 单减

收敛.

$\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)}$ $f(n) = \frac{1}{(n+1)\ln(n+1)}$, $\lim_{A \rightarrow \infty} \int_0^A \frac{dn}{(n+1)\ln(n+1)} = \lim_{A \rightarrow \infty} \ln \ln(A+1)$, 无界

(4) $\lim_{n \rightarrow \infty} \frac{\sqrt{n}-1}{n} = 0$, $f(n) = \frac{\sqrt{n}-1}{n}$, $f'(n) = \frac{\frac{n}{2\sqrt{n}} - \sqrt{n} + 1}{n^2} = \frac{1 - \frac{\sqrt{n}}{2}}{n^2}$, 故 $\frac{\sqrt{n}-1}{n}$ 先增后减

由于去掉有限项不影响敛散性, 故收敛

$\frac{\sqrt{n}-1}{n} > \frac{1}{\sqrt{n}} - 1$, 发散, 故条件收敛

(5) $u_n = \frac{2n+1}{n(n+1)} = \frac{1}{n} + \frac{1}{n+1}$, 易知 u_n 单减, 且 $\lim_{n \rightarrow \infty} u_n = 0$

$\sum_{n=1}^{\infty} \left(\frac{1}{n} + \frac{1}{n+1}\right) \geq \sum_{n=1}^{\infty} \frac{2}{n+1}$, 发散, 故条件收敛

(6) $u_n = \frac{n!}{3^{n^2}}$, $\frac{u_n}{u_{n-1}} = n \cdot 3^{(n-1)^2 - n^2} = n \cdot 3^{1-2n} = \frac{n}{3^{2n-1}} < 1$, 故 u_n 单减, 且 $\lim_{n \rightarrow \infty} u_n = 0$, 收敛

① $p > 0$
(2) $\lim_{n \rightarrow \infty} \frac{1}{(2n-1)^p} = 0$, 且 $\frac{1}{(2n-1)^p}$ 单减

故收敛.

$\sum_{n=1}^{\infty} |u_n| = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^p} \leq \sum_{n=1}^{\infty} \frac{1}{n^p}$, 若 $p > 1$, 则有 $\sum_{n=1}^{\infty} |u_n|$ 收敛, 原级数绝对收敛

$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^p} \geq \sum_{n=2}^{\infty} \frac{1}{(2n)^p}$, 若 $0 < p < 1$, 则有 $\sum_{n=1}^{\infty} |u_n|$ 发散, 原级数条件收敛

② $p = 0$, 易知发散

③ $p < 0$, $\lim_{n \rightarrow \infty} u_n \neq 0$, 发散.

$$\sum_{n=1}^{\infty} \frac{n!}{3^{n^2}}, \quad \lim_{n \rightarrow \infty} \frac{u_n}{u_{n-1}} = 0, \quad \text{故绝对收敛}$$

$$(7) \quad \sum_{n=2}^{\infty} \frac{1}{n} \sin \frac{2}{n} \cdot (-1)^n, \quad u_n = \frac{1}{n} \sin \frac{2}{n}, \text{单减, 且 } \lim_{n \rightarrow \infty} u_n = 0, \text{故收敛}$$

$$\frac{1}{n} \sin \frac{2}{n} \leq \frac{2}{n^2}, \text{绝对收敛}$$

$$(8) \quad u_n = \tan \frac{\varphi}{n}, \quad \varphi = 0, u_n \equiv 0, \text{收敛}; \varphi > 0 \text{时, 单减}, \varphi < 0 \text{时, 单增}, \text{故 } u_n \text{ 单调}$$

$$\lim_{n \rightarrow \infty} u_n = 0, \text{故收敛}$$

$$u_n \geq \frac{\varphi}{n}, \text{发散,}$$

$$\text{故条件收敛}$$

$$(9) \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^t (\ln n)^s} \quad (t > 0, s > 0);$$

$$(10) \quad \sum_{n=1}^{\infty} \sin(\pi \sqrt{n^2 + 1}) \quad (\text{提示: } \sin(\pi \sqrt{n^2 + 1}) = \sin[\pi(\sqrt{n^2 + 1} + n) - n\pi]);$$

$$(9) \quad u_n = \frac{1}{n^t (\ln n)^s}, \quad \lim_{n \rightarrow \infty} u_n = 0, \quad u_n \text{ 单减}$$

$$\text{故收敛}$$

$$\textcircled{1} t > 1 \text{时}, \quad \frac{1}{n^t (\ln n)^s} \leq \frac{1}{n^t}, \text{收敛, 绝对收敛}$$

$$\textcircled{2} t = 1 \text{时, 易知当 } s > 1 \text{时发散, 条件收敛}$$

$$s < 1 \text{时收敛, 绝对收敛}$$

$$\textcircled{3} 0 < t < 1 \text{时, } \begin{cases} s \geq 1, & \frac{1}{n^t (\ln n)^s} > \frac{1}{n (\ln n)^s}, \text{则有级数条件收敛} \\ 0 < s < 1, & \text{若此时 } s + t \leq 1, \frac{1}{n^t (\ln n)^s} \geq \frac{1}{n^{s+t}}, \text{发散, 条件收敛} \end{cases}$$

$$\downarrow$$

$$\text{一定存在 } N > 0, n > N \text{时 } (\ln n)^s < n^{1-t}, \quad \frac{1}{n^t (\ln n)^s} > \frac{1}{n}, \text{发散}$$

$$(10) \quad \sin(2\sqrt{n^2+1}) = \sin[(\lambda(\sqrt{n^2+1}+n) - n\lambda)] = \sin[\lambda(\sqrt{n^2+1}+n)] \cos n\lambda - \cos[\lambda(\sqrt{n^2+1}+n)] \sin n\lambda \\ = (-1)^n \sin[\lambda(\sqrt{n^2+1}+n)]$$

$$u_n = \sin \lambda(\sqrt{n^2+1}+n) = \sin \lambda(\sqrt{n^2+1}-n) = \sin \frac{\lambda}{\sqrt{n^2+1}+n}, \quad \text{显然 } \frac{\lambda}{\sqrt{n^2+1}+n} \text{ 在同一单调区间内}$$

故有 u_n 单调, 且 $\lim_{n \rightarrow \infty} u_n = 0$, 故收敛

$$\text{而 } \sin\left(\frac{\lambda}{n+\sqrt{n^2+1}}\right), \quad \sqrt{n^2+1} = n \cdot \sqrt{1+\frac{1}{n^2}} = n \cdot \left(1 + \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right)\right) = n + \frac{1}{2n} + o\left(\frac{1}{n}\right) \\ \therefore \frac{\lambda}{n+\sqrt{n^2+1}} = (\sqrt{n^2+1}-n)\lambda = \left[\frac{1}{2n} + o\left(\frac{1}{n}\right)\right]\lambda$$

$$\text{取 } b_n = \frac{1}{n}, \quad \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{\lambda}{n+\sqrt{n^2+1}}\right)}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{2n} + o\left(\frac{1}{n}\right)\right)}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2n} + o\left(\frac{1}{n}\right)}{\frac{1}{n}} = \frac{1}{2} + o(1)$$

故与 $\frac{1}{n}$ 敛散性相同,

条件收敛

$$(11) \quad \sum_{n=1}^{\infty} (-1)^n \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} \quad \left(\text{提示: 令 } u_n = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)}, \text{ 利用 } \frac{n+1}{n} < \frac{n}{n-1} \right. \\ \left. (n > 1), \text{ 可证 } u_n < \frac{1}{u_n} \cdot \frac{1}{2n}, \text{ 即 } u_n^2 < \frac{1}{2n}, u_n < \frac{1}{\sqrt{2n}}, \text{ 故当 } n \rightarrow \infty \text{ 时 } u_n \rightarrow 0, \text{ 再} \right. \\ \left. \text{利用 } \frac{n}{n+1} > \frac{n-1}{n} \text{ 可证 } u_n > \frac{1}{2u_n} \cdot \frac{1}{2n}, \text{ 即得 } u_n > \frac{1}{2} \cdot \frac{1}{\sqrt{n}}, \text{ 从而推出 } \sum_{n=1}^{\infty} u_n \text{ 发散} \right).$$

$$(11) \quad u_n = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} \quad \text{易知 } u_n \text{ 单调且 } \lim_{n \rightarrow \infty} u_n = 0, \text{ 故级数收敛}$$

$$\frac{n+1}{n} < \frac{n}{n-1} \quad u_n = \frac{1}{2n+1} \cdot \frac{(2n+1)!!}{2n!!} < \frac{2n!!}{(2n-1)!!} \cdot \frac{1}{2n+1} = \frac{1}{u_n(2n+1)}, \quad \therefore u_n^2 < \frac{1}{2n+1},$$

$$u_n < \frac{1}{\sqrt{2n+1}},$$

$$\text{又有 } \frac{n}{n+1} > \frac{n-1}{n}, \quad u_n = \frac{(2n-1)!!}{2n!!} > \frac{(2n-2)!!}{(2n-1)!!} = \frac{1}{2n} \cdot \frac{1}{u_n}, \quad u_n^2 > \frac{1}{2n}, \\ \therefore u_n > \frac{1}{\sqrt{2n}}$$

$$\therefore \frac{1}{\sqrt{2n}} < u_n < \frac{1}{\sqrt{2n+1}}$$

$$\text{而 } \sum_{n=1}^{\infty} \frac{1}{\sqrt{2n}} \text{ 发散,}$$

故条件收敛

2. 已知级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 证明级数 $\sum_{n=1}^{\infty} \frac{u_n}{n^p}$ ($p > 0$) 与 $\sum_{n=1}^{\infty} \frac{n}{n+1} u_n$ 均收敛.

3. 证明级数 $\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^p}$ ($0 < \varphi < 2\pi$) 当 $p > 1$ 时绝对收敛, 当 $0 < p \leq 1$ 时条件收敛.

4. 研究级数 $\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^p} \left(1 + \frac{1}{n}\right)^n$ ($0 < \varphi < 2\pi, p > 0$) 的敛散性.

5. 形如 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 的级数称作狄利克雷级数. 证明它有下列性质: 若级数 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛

(发散), 那么当 $x > x_0$ ($x < x_0$) 时, 级数 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 也收敛(发散).

6. 设级数 $\sum_{n=1}^{\infty} u_n$ 绝对收敛, 证明级数 $\sum_{n=1}^{\infty} \frac{2n-1}{n} u_n$ 也绝对收敛.

2. $\sum_{n=1}^{\infty} u_n$ 收敛, 则部分和序列有界

而 $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ 由狄利克雷判别法, 级数收敛
且单减

$\frac{n}{n+1}$ 单减且 $\rightarrow 1$, 而 $\sum_{n=1}^{\infty} u_n$ 收敛, 由阿贝尔判别法, 级数收敛

3. $p > 1$ 时, $\sum_{n=1}^{\infty} \left| \frac{\cos n\varphi}{n^p} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^p}$, 绝对收敛

$0 < p \leq 1$ 时,

$$\therefore \left| \frac{\cos n\varphi}{n^p} \right| \geq \frac{\cos^2 n\varphi}{n^p} = \frac{1 + \cos 2n\varphi}{2n^p}, \quad \frac{1}{2n^p} \text{ 发散, } \frac{\cos 2n\varphi}{2n^p} \text{ 收敛,}$$

故绝对值后发散

而 $\frac{\cos n\varphi}{n^p}$ 中, $\frac{1}{n^p} \downarrow$ 且 $\rightarrow 0$. 故原级数收敛

4. 易知 $(1 + \frac{1}{n})^n$ 单增且 $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$.

故 $(1 + \frac{1}{n})^n < e$.

$$\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^p} (1 + \frac{1}{n})^n < \sum_{n=1}^{\infty} \frac{e \cos n\varphi}{n^p}, \text{ 故级数收敛}$$

5. $\because n \in \mathbb{N}^*, \therefore \frac{1}{n^x}$ 随 x 单减

若 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛, 则有 $\forall \varepsilon > 0, \exists N > 0$, 当 $n > N$ 时, $\left| \sum_{k=n+1}^{n+p} \frac{a_k}{n^{x_0}} \right| < \varepsilon, \forall p > 1$

则有

$$\left| \sum_{k=n+1}^{n+p} \frac{a_k}{n^x} \right| < \left| \sum_{k=n+1}^{n+p} \frac{a_k}{n^{x_0}} \right| < \varepsilon \quad \text{错误, } a_n \text{ 正负不定, 大小关系不确定, 绝对值在外侧}$$

若 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 发散, 设 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ ($x < x_0$) 收敛, 由上分析 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛, 矛盾, 故 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 发散

6. $\left|\frac{2n+1}{n}\right| = 2 - \frac{1}{n}$ 单增且有 $\lim_{n \rightarrow \infty} \frac{2n+1}{n} = 2$.
 故 $\sum_{n=1}^{\infty} \frac{2n+1}{n} |u_n|$ 收敛 (阿贝尔判别法)
 故绝对收敛

7. 证明收敛级数 $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ 重排后的级数

$$1 + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{4k-3}} + \frac{1}{\sqrt{4k-1}} - \frac{1}{\sqrt{2k}} + \dots$$

发散. (提示: 先证 $\sum_{k=1}^{\infty} v_k$ 发散, 其中 $v_k = \frac{1}{\sqrt{4k-3}} + \frac{1}{\sqrt{4k-1}} - \frac{1}{\sqrt{2k}}.$)

$$\text{令 } v_k = \frac{1}{\sqrt{4k-3}} + \frac{1}{\sqrt{4k-1}} - \frac{1}{\sqrt{2k}}$$

则原级数为 $\sum_{k=1}^{\infty} v_k$,

$$v_k > \frac{2}{\sqrt{4k}} - \frac{1}{\sqrt{2k}} = \frac{1}{\sqrt{2k}}, \quad \text{而级数 } \sum_{k=1}^{\infty} \frac{1}{\sqrt{2k}} \text{ 发散,}$$

故 $\sum_{k=1}^{\infty} v_k$ 发散

设原级数与 $\sum_{k=1}^{\infty} v_k$ 的前 n 项和为

$$S_n, \quad \sigma_n$$

$$\text{则有 } \sigma_n = S_{3n}, \quad \lim_{n \rightarrow \infty} S_{3n} = \infty$$

$$\text{而 } S_{3n+1} = S_{3n} + \frac{1}{\sqrt{4n+1}}, \quad \lim_{n \rightarrow \infty} S_{3n+1} = \infty$$

$$\text{同理 } \lim_{n \rightarrow \infty} S_{3n+2} = \infty$$

$$\text{故 } \lim_{n \rightarrow \infty} S_n = \infty.$$

故原级数发散